

Exploring Display Parameters Associated with Cybersickness Using Electroencephalography

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ABSTRACT

VR offers immersive experiences but often causes cybersickness, hindering application. Here, we analyzed electroencephalographic (EEG) data for VR-induced discomfort associated with display parameters, as previous studies have not thoroughly examined the neural impact of individual display settings. Increased discomfort correlates with higher low-frequency and lower mid-frequency EEG signal power, indicating changes in autonomic activity and cognition. Our findings reveal the complex interplay between the display settings and EEG patterns. Optimizing the settings can mitigate discomfort and improve VR experiences, promoting wider application of VR technology.

Index Terms: VR-induced discomfort, Cybersickness, Electroencephalography (EEG), Display parameters

1 INTRODUCTION

Cybersickness is a discomforting experience induced by prolonged exposure to VR environments, accompanied by physical symptoms such as nausea, headache, eye fatigue, and vomiting [1, 9, 4]. Cybersickness is one of the major barriers to the spread of the use of this technology. The mechanism of the occurrence of cybersickness has been attributed to the discrepancy between the visual and vestibular sensation [6]. Traditionally, the simulator sickness questionnaire, a subjective measure, has been commonly used to assess cybersickness [2]. However, objective assessment methods are being developed as well.

Among the objective assessment methods, electroencephalography (EEG), which measures the electrical activity in the brain on the scalp, has recently attracted attention as a promising method to capture the mechanisms underlying the development of VR-induced discomfort leading to cybersickness. For example, Wibawa et al. (2019) reported EEG signal patterns of the occurrence of VR-induced discomfort by having participants play the 3-D video game “Mirror Edge” [9]. In Krokos et al. (2022), participants experienced a VR video content of navigating through 3-D space and used EEG to capture the characteristic patterns of VR-induced discomfort [4]. These studies provide important insights into the quantitative evaluation of the initiation of cybersickness, that is, VR-induced discomfort.

Although, as mentioned above, the EEG patterns during VR-induced discomfort are being empirically studied, the characteristics of EEG signals for VR-induced discomfort which are associated with display parameters remain unclear. The degree to which

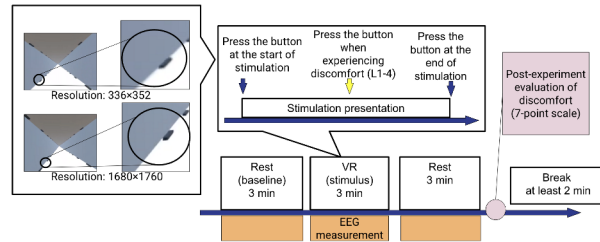


Figure 1: Experimental procedure.

Table 1: Experimental conditions of display parameters.

Experiment day	Jittering frequency [Hz]	Resolution [pixels]	Refresh rate [Hz]
Day 1	0	1680×1760	120
	20	1680×1760	120
	40	1680×1760	120
	60	1680×1760	120
Day 2	20	1680×1760	72
	20	1680×1760	120
	20	336×352	72
	20	336×352	120

display parameters, such as jittering frequency, resolution, and refresh rate, relate to VR-induced discomfort, can only be investigated by specifically isolating each of them during experiment as a factor affecting VR-induced discomfort; the previous studies were conducted with these factors mixed together. Though recent advancements in VR technology have gradually improved visual fidelity, frequent occurrences of small misalignments in the visual stimuli, such as jittering, can still cause discomfort [11]. In order to explore the detailed characteristics of EEG signals for VR-induced discomfort, we analyzed EEG data under the conditions in which each of the display parameters was controlled separately in the VR environment.

2 MATERIALS AND METHODS

Eighteen participants (ten females and eight males, mean age: 20.4 ± 2.0) participated in the experiment. They were informed of the study's purpose and procedures beforehand, and written informed consent was obtained.

Participants experienced a VR content (Fig. 1) using Meta Quest 3 (Meta Inc.) while sitting on a comfortable chair. The EEG data were collected before, during, and after the VR experience. The experimental conditions of the VR content were split into eight combinations of the following display parameters: the frequency of jittering noise with a distance of 0.1 (unit in Unity), resolution, and refresh rate (Tab. 1). The jittering noise here represents the mean frequency fluctuations of the jitter, which followed a normal distribution. The experiment was conducted over two days (Day 1 and Day 2), where the order of the four experimental conditions in

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each day was counterbalanced among participants. In each experimental condition, the VR experience was presented for a duration of 3 min. Participants reported their level of discomfort during the VR experience in real time using a controller. The discomfort levels were classified into four categories: L1, L2, L3, and L4, each corresponding to no discomfort, slight discomfort, moderate discomfort, and severe discomfort, respectively. This classification allowed for a nuanced understanding of the participants' subjective experiences throughout the VR immersion.

EEG data were recorded with a 64-channel actiCHamp system (Brain Products GmbH) and preprocessed using EEGLAB with MATLAB (Mathworks Inc.). Preprocessing steps included line noise removal, filtering, channel interpolation, and independent component analysis. Subsequently, spectral analysis was performed to analyze EEG signal power in different frequency bands: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–25 Hz), and gamma (25–45 Hz) bands [7, 4, 8]. Particularly, theta wave fluctuations have been reported to be associated with VR-induced discomfort [7, 4, 8]. For statistical analysis, repeated measures analysis of variance with a factor of the frequency band was used.

3 RESULTS

Fig. 2 shows the comparison of EEG power spectra across frequency conditions with varying levels of discomfort, where the main effect of the frequency band was significant ($F(3,51) = 4.2$, $p < 0.05$) with increased power in the low-frequency range (delta and theta bands) and decreased power in the mid-frequency range (alpha and beta bands) as discomfort increased. The current analysis is done on EEG data related with all of the display parameters. However, as of now, it was found that overall, the frequency of jittering noise tended to have relatively higher impact on discomfort than the other display parameters.

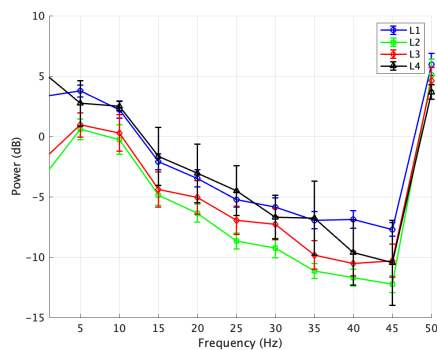


Figure 2: EEG power spectra for discomfort levels.

4 DISCUSSION

The findings in this study suggest that the mechanisms underlying VR-induced discomfort leading to cybersickness are influenced by complex factors beyond simple changes in brain activity. Specifically, the effects of jittering noise and refresh rate changes on EEG signals were found to be limited, indicating that these factors alone are unlikely to be the main causes of discomfort. The observed increase in low-frequency power (delta and theta bands) with increasing discomfort levels reflects changes in autonomic nervous system activity and arousal levels [10, 3], potentially corresponding to relaxation and decreased attention. In contrast, reduction in mid-frequency power (alpha and beta bands) is associated with increased cognitive load and altered sensory integration [5, 12], highlighting the negative impact of discomfort on the regulation of sensory input and cognitive processing.

Future research should adopt a more comprehensive approach, integrating environmental factors, individual susceptibility, and other physiological indicators (e.g., heart rate, skin conductance). Additionally, real-time detection of discomfort which leads to cybersickness and the development of appropriate interventions will be crucial to the enhancement of the quality of VR experiences. Our findings provide valuable insights which could help with designing VR contents to reduce cybersickness, offering actionable guidelines for creating much comfortable and engaging VR environments.

5 CONCLUSION

We analyzed EEG data for VR-induced discomfort leading to cybersickness associated with display parameters. Our results show that increased discomfort correlates with higher low-frequency EEG signal power and lower mid-frequency power, indicating changes in autonomic activity and cognitive processing. The current analysis includes EEG data for all of the display parameters, however, as of now, frequency of jittering noise was found to have relatively higher impact on discomfort than the other display parameters. By focusing on these factors, we can enhance user comfort and promote wider application of VR.

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